GATE

Computer Science & Information Technology

Compiler design

DPP: 2

Parsing

Q1 Identify the correct statement when LR parsers compared using any particular grammar.

(A) If LR (0) has no conflict then SLR (1) may have the conflicts

(B) If SLR (1) has no conflict then LR (0) never contain any conflict

(C) If LALR (1) has no conflict then CLR (1) may contain any conflict

(D) If CLR (1) has no conflict then LALR (1) may have the conflicts

Q2 Consider the following grammar

S → Aa | bAc | dc

A → d

The above grammar is ___________.

(A) SLR(1)

(B) LALR(1) but not SLR(1)

(C) CLR(1) but not LALR(1)

(D) None of these

Q3 Which of the following sets could result SR conflict in LALR (1)?

(A) A → a.b, {b}
B → a.,{a}

(B) A →a.a, {a}
B → a., {b}

(C) A → b.a, {b}
B → b., {a}

(D) A → b.b,{a}
B → b., {a}

Q4 Consider the following operator grammar

E → AaBcD

A → bA|e

B → bB|e

D → eD|g

Which of the following precedence relation is incorrect from the above grammar? Assume x <· y is used to represent x is having less precedence than y and in the expression x appears first and then y appears next.

(A) a <· b (B) c <· g

(C) a <· e (D) c <· b

Q5 Consider the following grammar,

S→A+B | A-B | A | B

A → a | b

B → b | c

Which of the following statement(s) is/are correct?

S1: LL(1) can parse all the strings that are generated by the grammar.

S2: LR(1) can parse all the strings that are generated by the grammar.

(A) Neither S1 nor S2

(B) Only S1

(C) Only S2

(D) Both S1 as well as S2

Q6 Consider the following grammar,

S → AA

A → aA

A → b

Let X is the number of states in the LR(0) parsig table and Y is the number of states in the CLR(1) parsing table for the above grammar, then |X-Y| is_______

Q7 Consider the following augmented grammar,

S'→ S

S → L=R | R

L → *R | id

R → L

goto(Closure(S→L= .R),) contains exactly __________ items.

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Answer Key

Q1	(D)
Q2	(B)
Q3	(C)
Q4	(D)
Q5	(A)
Q6	3
Q7	4

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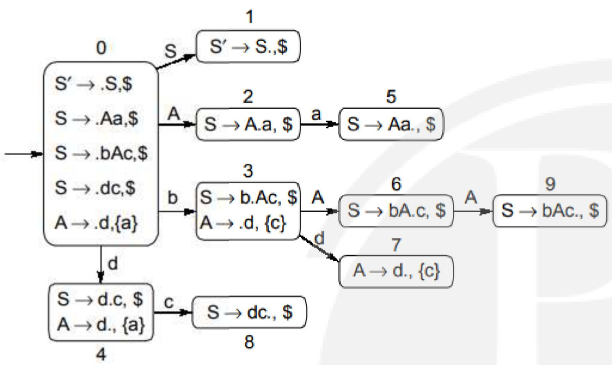
Hints & Solutions

Q1 Text Solution:

If grammar is LR (0) then it is also SLR (1). It implies if no conflict present in LR (0) then SLR (1) has
no conflict.
If CLR (1) has no conflict then LALR (1) may have SR conflict.
Hence option (d) is correct

Q2 Text Solution:



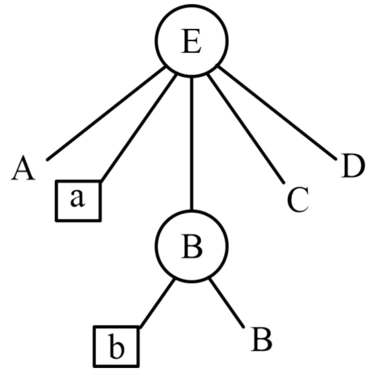
IN state 4; SR conflict occurs in SLR (1) c ∈ Follow (A)
IN State 4; No SR conflict in LALR (1) c ∉ {a}
∴ Given grammar is LALR (1) but not SLR (1).

Q3 Text Solution:

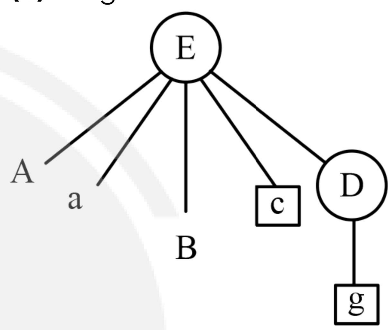
A → b.a, {b}
B → b., {a}
a belongs to set of look-a-heads of a B. Shift reduced conflict occurs in option (c)

Q4 Text Solution:

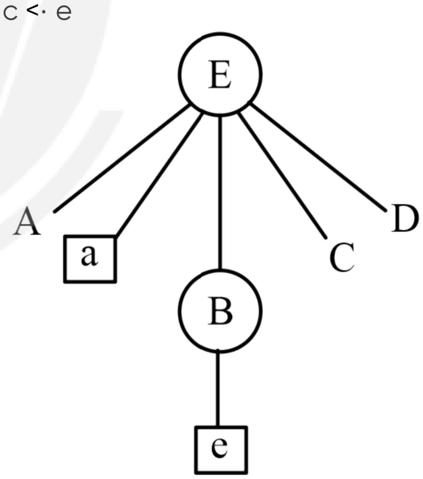
E → AaBcD
A → bA|e
B → bB|e
D → eD|g
(a) a <· b



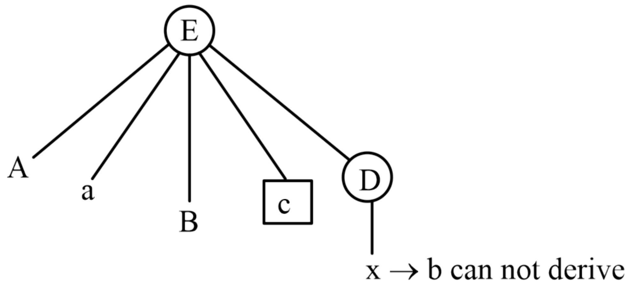
⇒a<·b is correct
(b) c<·g



⇒c<·g is correct
(c) c <· e



⇒a <· e is correct
(d) c <· b



⇒c <· b is incorrect precedence relation.

Q5 Text Solution:

S1: LL(1) can parse all the strings that are generated by the grammar. **INCORRECT**

S2: LR(1) can parse all the strings that are generated by the grammar. **INCORRECT**

Q6 Text Solution:

Number of states in LR(0) = 7

Number of states in CLR(1) = 10

|7 - 10| = 3

Q7 Text Solution:

Number of items = 4